

Size

The imbalance size expressed as a percentage of ADV. It is expressed as a decimal, that is, an imbalance size of 30% ADV is expressed as 0.30.

$$Size = \frac{Q}{ADV} \quad (5.11)$$

POV Rate

The percentage of volume rate (POV) is computed from imbalance and period volume. It is a proxy for trading strategy. It is important to note that percentage of volume is expressed as a decimal.

$$POV = \frac{Q}{V(t)} \quad (5.12)$$

Cost

Cost is defined as the difference between average execution price and the first price (expressed as a fraction of the initial price). It follows the definition of trading cost used in the implementation shortfall methodology (Perold, 1988). Here we compute cost as the logarithmic price change between average execution price and arrival price. We use the VWAP price over the interval as our proxy for average execution price. This calculation is as follows:

$$Cost = \ln\left(\frac{VWAP}{P_0}\right) \cdot Side \cdot 10^4 bp \quad (5.13)$$

Additional Insight: we discuss the expanded implementation shortfall measure introduced by Wagner (1991) in Chapter 3, Transaction Cost Analysis. In Chapter 6, Price Volatility, we discuss techniques to estimate forward looking volatility. In Chapter 7, Advanced Algorithmic Forecasting Techniques, we discuss alternative methodologies for estimating daily volume. These include using various time periods, the mean vs. median, as well as advanced statistical measures such as autoregressive moving averages (ARMA).

Estimating Model Parameters

Estimation of the parameters for the complete I-Star model requires non-linear estimation techniques such as non-linear least squares, maximum likelihood, generalized method of moments, etc. In Chapter 4, Market Impact Models, we discuss three techniques including a two-step process, a guesstimate, and non-linear regression analysis.